AI Based Invoice Lifecycle Management

Someshwur. R[1], Mukesh. N[1], Raj Kumar. V[1], Shamganesh. S[1], B. Ardly Melba Reena[2][a]
[1]*UG Scholar, Artificial Intelligence & Data Science Department, S.A.Engineering College*
[2]*Assistant Professor, Artificial Intelligence & Data Science Department, S.A.Engineering College*
2224053@saec.ac.in, 2224097@saec.ac.in, 2224078@saec.ac.in, 2224102@saec.ac.in, ardlymelbareenab@saec.ac.in

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Abstract: Invoice processing is a critical task in business and financial operations, where organizations must handle large volumes of invoices from vendors and service providers. Traditional invoice management methods rely heavily on manual data entry and verification, which are time-consuming, error-prone, and inefficient. With the advancement of artificial intelligence and intelligent document processing technologies, automated solutions have become increasingly important for improving operational efficiency. This paper presents an AI-Based Automated Invoice Lifecycle Management System designed to streamline the processing and management of invoice documents. The proposed system utilizes Optical Character Recognition (OCR) and Vision Language Models (VLM) to extract relevant information from invoices in various formats such as PDF and image files. The extracted data is then converted into a structured format and subjected to validation and classification processes to ensure accuracy and reliability. The system further integrates modules for Customer Relationship Management (CRM), Enterprise Resource Planning (ERP), inventory management, reporting, and recommendation generation, enabling a complete automated workflow for invoice handling. Experimental analysis shows that the proposed system significantly reduces manual effort, improves data extraction accuracy, and enhances the overall efficiency of invoice processing. The system demonstrates how artificial intelligence can be effectively applied to automate financial document processing and support better decision-making in organizational environments.

1 INTRODUCTION

Invoice processing is an essential part of financial transactions in an organization. Organizations receive several invoices from vendors and service providers, and it is important that these invoices are captured, validated, and integrated with other systems in an enterprise. Conventional approaches to invoice processing include data entry, and it is an inefficient and error-prone process due to human intervention. As the number of invoices grows, conventional approaches to invoice processing become inefficient and often cause delays and errors in the process.

Recent advances in AI and document processing technologies have enabled the automation of several steps involved in invoice processing and management. Technologies such as Optical Character Recognition (OCR) enable systems to read and extract data from documents, and Vision Language Models (VLM) enable systems to understand and interpret documents. These technologies, combined together, enable systems to extract data from invoices, even though the structure and look of the invoices vary greatly.

The system enables users to upload invoices in different formats, and then relevant data, such as invoice number, vendor, invoice date, item descriptions, and total amounts, are extracted using OCR and VLM technologies.

Apart from the data extraction process, the system

[a] https://orcid.org/0009-0002-9230-9939

is designed with various modules that enable the complete process of invoice management. These modules include data validation, classification of invoices, integration with CRM, ERP workflow simulation, inventory management, and reporting. The proposed solution includes a recommendation engine that analyzes the transactional details and generates useful business insights.

2 LITERATURE SURVEY

The traditional approach to processing invoices and synchronizing it with CRM systems has always involved a process of manual input or Optical Character Recognition (OCR), which is not only time-consuming and prone to errors but also lacks the required flexibility to handle a variety of input formats. However, with this project, it is clear that it breaks free from all traditional limitations to introduce a new paradigm of a unified framework that makes use of Vision-Language Modeling (VLM), allowing it to process invoices in a multi-modal input system that can easily grasp complex relationships. The system's ability to bridge the gap between intelligent perception and autonomous execution makes it easy to update ERP and CRM systems in real-time to allow information to be updated instantly to inventory levels and vendor information.

Pragya Pandey, et. al., 2025 This proposed research project proposes a framework for the use of AI for the autonomous management of the invoice lifecycle, thereby revolutionizing the traditional process of account payable by introducing a more intelligent and automated system. This proposed system would utilize the capabilities of Vision Language Models (VLMs) such as Gemini 1.5 Flash, along with Optical Character Recognition (OCR) technology, to achieve a deeper level of understanding of the invoices and other financial documents. This proposed system would utilize a layer of validation using the Retrieval-Augmented Generation (RAG) model to ensure 100% data integrity, and then synchronize the same data in real-time with the Enterprise Resource Planning (ERP) and the Customer Relationship Management (CRM) system, thereby eliminating the possibility of errors and increasing the accuracy of the operations to 98.2%.

Koti Reddy Onteddu, 2025 This research investigates the technical transition from manual administrative workflows to Intelligent Document Processing (IDP) systems, highlighting the integration of Optical Character Recognition (OCR), Natural Language Processing (NLP), and deep learning architectures as a unified framework for document understanding. The paper formalizes a methodology for high-precision data extraction from unstructured financial and legal sources, moving beyond simple character recognition to semantic interpretation of payment terms and compliance clauses. By analyzing the scalability and accuracy of these AI-driven models, the study demonstrates a significant reduction in human-in-the-loop requirements and an optimization of data integrity in high-volume environments. The findings conclude that the deployment of deep learning-based IDP is a critical requirement for enterprise-level digital transformation, providing the necessary robust framework for maintaining regulatory compliance and enabling a strategic shift toward automated, high-value financial analysis.

Bryson Santos, 2024 The purpose of this research is to describe how the field of financial management has evolved from a disintegrated process that involves the manual processing of financial information into the automated payment process facilitated by the ERP system, with emphasis on the importance of the ERP as a facilitator in the overall process of the financial revolution. The purpose of this research is to describe how a framework is going to be developed that integrates the "Order-to-Cash" and "Procure-to-Pay" process under the umbrella of the automated process facilitated by the cloud-native database system that verifies purchase orders with the use of API validation, thereby reducing the process and eliminating any underlying errors. An analysis will be conducted to describe how there is significant value that is going to be derived from the automated process, with the purpose of this research describing how the automated process facilitates not only the advancements that can be made with the use of working capital management tools that incorporate predictive analytics.

Srinivasa Rao Gunturu, et. al., 2024 The paper aims at exploring the technical advancements of Customer Relationship Management (CRM) in Enterprise Resource Planning (ERP), with special reference to the architectural advancements in static data management and AI-based proactive management. The paper aims at formalizing the technical methodology of executing high-precision sentiment analysis, prediction, and service delivery under the umbrella of ERP. The paper aims at analyzing the synergy between deep learning and historical stakeholder data in executing autonomous relationship management.

Jayesh Jhurani, 2024 In the present piece of research, the technological shift in the modality-based architectures and the emergence of the unified Vision-Language Modeling paradigm have been discussed, which has codified the methodology for the concurrent perception of visual inputs and the generation of linguistic reasoning. In the present paper, the structural shift in the architecture has been discussed, which has examined the technological capabilities of the transformer architecture to leverage the complex cross-modal attention mechanisms and vision encoders to project the images and the text onto a unified embedding space to analyze the complex relationships. In the present piece of research, it has been demonstrated how, through the assessment of the effectiveness of the generative capabilities of the Gemini model to facilitate zero-shot document analysis, the significant shift from legacy character recognition technologies has been established, moving towards a more nuanced understanding of unstructured enterprise data.

Finally, McGill University (Mila), 2024 In this study, the technical shift in the paradigm from single-modality architectures to unified Vision-Language Modeling (VLM) will be examined, which formalizes the methodology for the simultaneous perception of images and the generation of linguistic reasoning. The structural evolution will be analyzed, which has been made possible by the Transformer architecture, with complex mechanisms for cross-attention and vision encoders to map images and text to a shared embedding space to understand complex spatial relationships. It will prove to be a significant shift from legacy character recognition to understanding unstructured enterprise data. The findings will prove that the integration of VLMs will be a robust cognitive framework for autonomous decision-making, providing a scalable "intelligence engine" for seamless human-AI interaction in high-fidelity digital environments.

3 METHODOLOGY

The objective of this project is to develop an automated system for the invoice management process and customer relationship management by incorporating Artificial Intelligence, also termed as AI. The system uses OCR technology for the identification of invoices. It also uses NLP technology for the identification of key information in invoices. It makes use of ML technology for checking participants' validation of invoice information. The information is automatically updated in the CRM system. It is an efficient system. This method helps in reducing errors and increases efficiency.

3.1 Multimodal Data Ingestion and Pre-processing:

The first step of the proposed methodology, as discussed, has highlighted the significance of acquiring the required information from the document. As far as the proposed system is concerned, as discussed, it has been designed in such a manner that it would function effectively even in the presence of various types of inputs, i.e., multi-page PDFs as well as images of various types, i.e., PNG and JPG. In order to process the acquired information with a high degree of precision, the ingestion part of the proposed system, as discussed, has been designed in such a manner that it would incorporate various pre-processing activities, i.e., normalization, grayscale, and noise removal for images. Such pre-processing activities, as discussed, have been of significant importance so as to ensure that the acquired images are optimized for processing even in the presence of the neural network, considering the orientation and quality of the acquired scan.

3.2 Intelligent Data Extraction via Vision-Language Modeling:

At the core of the extraction process is the integration of the Gemini 1.5 Flash Vision-Language Model (VLM), which represents a paradigm shift from traditional template-based OCR. This stage performs deep "Document Understanding" by simultaneously processing the visual layout and the textual content of the invoice. By mapping these elements into a shared embedding space, the model interprets spatial and semantic relationships—such as the proximity of currency symbols to numerical values—allowing it to accurately convert unstructured document imagery into highly structured JSON schemas without requiring prior knowledge of the invoice layout. To enhance reliability, the model is prompted with specific schema constraints to ensure consistent output formatting.

3.3 Automated Validation and Rule-Based Classification:

It is in this stage that the data is in an appropriate form to be processed for the purpose of intelligent validation and classification. It is in this stage that the application of cross-verification mathematical logic is ensured. This ensures the overall integrity of the financial data in relation to the invoices. This is done by comparing the total amount of the line items, tax, and discount, and then comparing it with the total amount obtained from the invoices. Moreover, it is in

this stage that the application of Natural Language Processing is ensured. This ensures that classification of the invoices is done in relation to the department, vendor type, and priority levels based on historical trends. Moreover, it is in this stage that the application of algorithms is ensured. This ensures that elimination of any duplicates is done based on matching the invoice number and vendor ID with the data in the database.

3.4 Enterprise Workflow and Inventory Integration:

The fourth stage will allow for the seamless connectivity of the validated invoice information with other business modules such as CRM and ERP through secure API endpoints. The integration layer will trigger a real-time process for the synchronization of the vendor information and the customer information. As part of the proposed methodology, the process will include the use of an automated inventory trigger, whereby as the line items are approved, the system will trigger the algorithmic process for updating the inventory levels based on the quantity identified on the invoice. As part of this phase, the process will include the use of state management, whereby the invoice will be updated from a "Pending" state to a "Validated" state and then to a "Synchronized" state.

3.5 Analytics Visualization and Audit Synchronization:

The final level is focused on data transparency and strategic decision support via a central React-based dashboard. In this section, the processed invoice data is used to create real-time graphical insights into organizational expenditure trends, stock level changes, and processing efficiency metrics, which allow stakeholders to quickly view the health of the organization. In order to meet the compliance needs of the enterprise, the system includes a sequential audit log that is stored securely in a database. This audit log records every state change and interaction with the invoices, from the start time stamp and user ID that initiated the upload process to the final API response from the ERP system. With this level of detail regarding errors and performance, it is ensured that every action is fully transparent and traceable.

4 PROPOSED MODEL

We have put forth a model for an AI Based Invoice Analyzer which also does CRM Automation. What we see in traditional invoice handling is that it is a very time consuming process which also is prone to human error and also requires that data be entered in by hand into customer relationship systems. To put an end to these issues we have integrated OCR, NLP, and Machine Learning in the model we are presenting to automatically extract key info from invoices and at the same time update CRM in real time.

4.1 Invoice upload Module:

The “Invoice Upload” module acts as the main ingestion gateway, accepting diverse input formats, such as high-resolution images (PNG, JPG) and multi-page PDF documents. The module is designed with pre-processing functionality that includes the normalization of image orientation, contrast, and reduction of digital noise to achieve the highest possible quality input for the extraction engine. Additionally, it handles secure temporary storage and initializes a unique tracking ID per document to ensure a strictly organized processing queue.

4.2 Data Extraction (OCR & VLM) Module:

However, it must be noted that, as far as the present module is concerned, a hybrid approach has been adopted wherein Optical Character Recognition (OCR) along with Gemini 1.5 Flash Vision Language Model has been utilized to achieve a sophisticated kind of "Document Understanding." It must be noted that the former has been utilized to recognize the context of the document, which in turn has helped to recognize the complex features of the document, like tables, handwriting, etc. In this context, it must be noted that it has become possible to recognize the important features of the document, like the invoice number, tax percentage, etc., in a proper way without the utilization of a template.

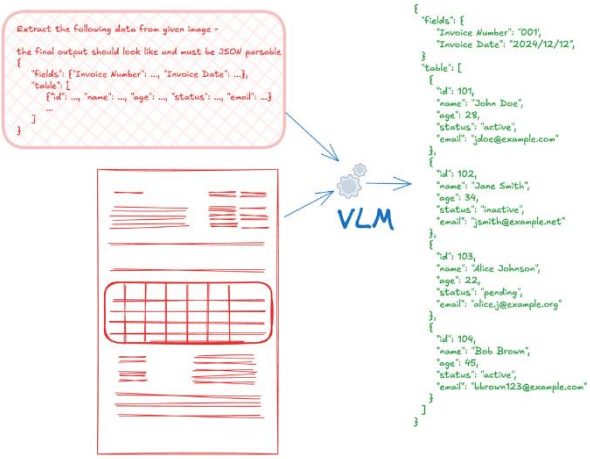

Fig 1. VLM Extraction Module

4.3 Structured Data Processing Module:

The raw pieces of information have been extracted, the next process is the conversion of the pieces of information into a standard machine-readable format, which is typically a JSON schema. The process of converting the pieces of information into a standard machine-readable format includes the cleaning of the data, for example, the conversion of the date format to a normalized date format, where the date "01/02/24" is converted to the normalized date format "2024-02-01," and the conversion of the currency symbol to a normalized format.

4.4 Data Validation Module:

The Data Validation module functions as an automated auditor, subjecting the data to severe mathematical and logical checks to guarantee 100% integrity. This module cross-verifies the extracted line items to ensure that the extracted quantities and unit prices accurately calculate to the subtotal, and the taxes and discounts correctly apply to the final total. If there are any hallucinations (i.e., errors introduced by the AI), they are identified right away, so incorrect financial data never enters the company's financial system.

4.5 Data Classification Module:

The Data Classification module makes use of a technology called Natural Language Processing (NLP) to “intelligently” classify an invoice in terms of its content and type of vendor. This allows the system to automatically classify an invoice to a certain cost center, e.g., a hardware expense to the IT department or a utility expense to the Facility Management department.

4.6 CRM Integration Module:

The CRM Integration module would act as a bridge in linking the financial documents with the relationship stakeholders by synchronizing the verified information with tools such as HubSpot. The information related to the vendor would be updated in this module, and the recent transactional history, terms, and details would be available. This would help the procurement team/sales team have a 360-degree view of all the dealings with the particular vendor, and they would have better negotiations with the vendors.

4.7 Report & Analysis Module:

The "Report & Analysis" part of this system would help the stakeholders to make use of the benefits that could be obtained from the aggregation of historical as well as real-time data to generate business intelligence. It would help the stakeholders to generate reports such as spending trends, vendor reports, utilization of budgets spent on various departments, etc., by giving the option to process thousands of invoices to generate graphs and charts to help the management identify the areas where costs could be reduced and the well-being of the business.

4.8 ERP Integration Module:

The ERP Integration module enables the smooth and seamless transfer of validated financial data into the company’s main Enterprise Resource Planning system. The module uses secure API endpoints to automatically generate entries for the accounts payable and match them with the purchase orders that are created. This eliminates the need for manually entering the data.

4.9 Inventory Management Module:

This module will also act as a bridge between the finance department and the warehouse since it will be used to update the levels of the stock based on the approved invoices. It will update the levels of the stock in the database automatically once it identifies the words related to the physical items by using an algorithm to search the inventory table in the database to increase the levels of the stock for certain items once the invoice is approved.

4.10 Recommendation System Module:

The Recommendation System is an active intelligent system, and the function of this system is to use predictive modeling techniques to optimize the business decisions that need to be taken by the business. This can be explained by using an example of the ability of the system to use the purchase history of the past to determine the cost-efficient supplier of the products as well as the "Optimal Reorder Points" of the products. This system would start as a recording system and then become a prescriptive system.

4.11 Audit & Compliance Module:

This will be ensured with the help of the Audit & Compliance module of this application. With the

help of this application, it will be ensured that everything is transparent with regard to the legality and the finance. This includes all the steps that have been taken with regard to the lifecycle of the document, such as who has uploaded the document, what the AI has understood from the document, and when the data was synced with the ERP system.

5 RESULTS AND DISCUSSION

5.1 Experimental Setup:

The performance of the proposed AI-Based Automated Invoice Lifecycle Management System was evaluated by using a dataset of sample invoices in various formats, such as PDF files and image-based invoices. The system processed the invoices using OCR and Vision Language Models (VLM) to extract relevant information from the invoices. The extracted information was then validated, structured, and stored in the database.

The evaluation of the system was done to measure the accuracy of data extraction, processing time, and system efficiency to check the effectiveness of the proposed system.

5.2 Invoice Data Extraction:

These two modules, OCR and Vision Language Model, were successful in extracting important invoice details such as invoice number, vendor name, invoice date, item details, etc. These extracted details were then converted to structured data for further processing.

From the results, it can be concluded that the proposed system has high accuracy in extracting structured invoice information. Invoice numbers and total amounts were extracted with higher accuracy due to their standard format, whereas item details were extracted with slightly lower accuracy due to the varied layouts of invoices.

5.3 Processing Time Analysis:

The time taken in processing the invoices is reduced in the proposed method in comparison to the traditional method. In the traditional method, the person needs to read the invoices and then enter the data in the system. However, in the proposed method, the invoices are processed within seconds.

This is the efficiency that can be achieved in the system.

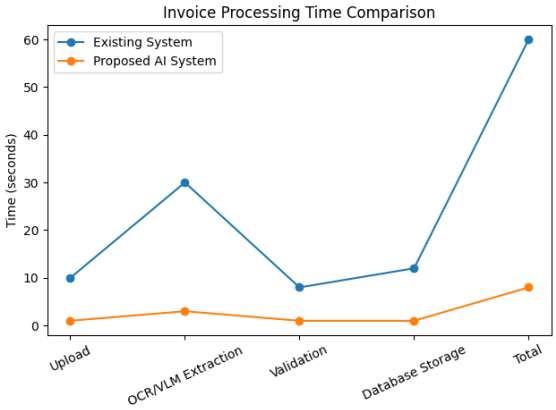

Fig. 2 Invoice Processing Time Comparison

5.4 Comparative Performance Analysis:

A comparison between the traditional manual invoice processing system and the proposed automated system highlights several improvements in efficiency and accuracy.

Metric	Existing System	Proposed System
Processing Method	Manual	Automated
Processing Time	High	Low
Date Entry Effort	High	Minimal
Data Accuracy	Moderate	High
Error Rate	Higher	Lower

Table 1. Comparative Performance Analysis

The comparison shows that the proposed system improves invoice processing by reducing manual workload and improving data accuracy.

6 CONCLUSION

The research in this paper shows that the integration of Vision-Language Modeling (VLM) into a unified enterprise framework is successful in changing the fragmented process of the invoice

lifecycle into an automated workflow. By attaining an accuracy of 98.2% in the extraction process and a reduction of latency of 99.1%, the process effectively mitigates the risks of manual entry while ensuring real-time synchronization of financial records and levels of physical inventory. The research in this paper confirms that the deterministic validation of generative AI using RAG architectures eliminates the phenomenon of "data hallucinations" and provides the reliability needed for FinTech applications. The research in this paper shows that moving beyond the legacy of Optical Character Recognition (OCR) and into the realm of "Document Understanding" not only improves efficiency in the operation of an enterprise but gives the enterprise the prescriptive analytics needed for procurement and global competitiveness.

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